Investigate Hotel Business using Data Visualization

Data Visualization Project Brief • Understanding Booking & Cancellation Behaviour

Tool	Jupyter Notebook (Python)	**Libraries**	Pandas, NumPy
Visualization	Matplotlib, Seaborn	**Dataset**	Hotel bookings, 2017–2019 (~119K rows × 29 cols)

In this project you will act as a data analyst for a hotel company. Business performance analysis is key to understanding customer behaviour: what drives guests to book, and what drives them to cancel. Using a large real-world booking dataset spanning 2017–2019, you will clean the data, then build clear visualisations to answer three business questions about hotel type, stay duration, and lead time. Finally you will turn your charts into a summary and business recommendations. Work through the stages below and answer every question with code and visualisations in your notebook.

> ***What to submit:*** *a single Jupyter Notebook (.ipynb) with your cleaning code, every chart, and your written answers, exported to PDF or HTML. Each chart must have a title, labelled axes, and a short caption explaining what it shows. Every insight you state must be visible in one of your own visualisations.*

STAGE 0 Problem Statement

1. In your own words, explain why understanding **customer booking behaviour** matters for a hotel business. What kinds of decisions could it improve?
2. This project is built around three **business questions**. Restate them clearly in your own words:

- Which hotel type do customers book most often?
- Does the length of stay affect the booking cancellation rate?
- Does lead time (the gap between booking and arrival) affect the cancellation rate?

1. State the **objective** of the project. What is the single main deliverable you are producing for the business?

STAGE 1 Data Preprocessing

1.1 Data Overview

1. Load the dataset. How many rows and columns does it have, and what period does it cover? Briefly describe the columns most relevant to the three business questions.

1.2 Data Assessment

Assess the data quality and decide how to handle each issue. Justify every decision.

1. Check for **missing values**. Which columns are affected (for example: company, agent, children, city)? For each, decide how to fill or handle it and explain what your chosen value implies about that guest.
2. Check for **duplicate rows**. Report your finding and the action taken.
3. Look for **inconsistent or unclear values** — for example, what does an 'Undefined' entry in the *meal* column mean? Propose a sensible way to group or recategorise such values.

4. Look for **anomalies**: e.g. negative or extreme values in the average daily rate (adr), and bookings that record no guests at all. How will you handle these rows, and why?

STAGE 2 Data Analysis

2.1 Monthly Booking Analysis by Hotel Type

1. Visualise the **share of bookings** between City Hotel and Resort Hotel. Which type is booked more often, and by roughly what percentage?
2. Plot the **number of bookings per month** for each hotel type. Which months are busiest and which are quietest?
3. What might explain the seasonal pattern you see? Relate it to holidays, travel habits, or business travel.
4. What operational or revenue decisions could a hotel make based on this monthly trend?

2.2 Impact of Stay Duration on Cancellation Rate

1. Calculate and visualise the **cancellation rate** for each hotel type. Which type is cancelled more often?
2. Plot how the cancellation rate changes as the **total length of stay** increases, for both hotel types. Describe the trend you observe.
3. Is the relationship between stay duration and cancellation positive, negative, or flat? Does it differ between the two hotel types?
4. Suggest at least two possible reasons **why** longer stays might be cancelled more (or less) often.

2.3 Impact of Lead Time on Cancellation Rate

1. Define **lead time** in your own words. Plot how the cancellation rate varies with lead time for each hotel type.
2. At what lead time is cancellation lowest? At what point is it highest, and for which hotel type?
3. Compare the two hotel types: does one stay more stable across lead times while the other rises sharply? What could explain the difference?

STAGE 3 Summary and Recommendations

Bring your findings together. For each of the three business questions, summarise what your visualisations showed and turn it into action.

1. Write a short **summary** of your key findings across all three analyses (hotel type popularity, stay duration, and lead time).
2. Based on **hotel type and seasonality**, give at least two recommendations — e.g. how to grow the less-popular hotel type, or how to capitalise on peak seasons.
3. Based on **stay duration and cancellations**, recommend policies that could reduce revenue lost to cancellations (consider cancellation terms, pricing, or minimum-stay rules).
4. Based on **lead time**, propose actions to reduce cancellations for far-ahead bookings (consider reminders, deposits, rescheduling, or pricing).
5. Which single recommendation do you believe would have the **biggest impact**, and why? Support it with evidence from your charts.

> ***Presentation tip:*** *Imagine you are presenting to a hotel manager who does not code. Make sure each chart tells one clear story, and that your recommendations follow directly from what the data shows — not from assumptions.*